

Problem 3 – Smart Student Grade Calculator 🎓

Difficulty

★ Beginner

Overview

A school wants to simplify the process of calculating students' final grades. Instead of calculating the average manually, teachers would like a smart assistant that computes the average, determines whether the student passed or failed, and assigns the appropriate letter grade.

Your task is to build a Student Grade Calculator that can process multiple students.

Mission (Python)

1. Store the grading system.

Use a dictionary or another suitable data structure to represent the grading scale.

Average Grade

90 – 100 A

80 – 89 B

70 – 79 C

60 – 69 D

Below 60 F

2. Make the application run forever.

For every student, ask for:

- Student name
 - Math grade
 - Science grade
 - English grade
-

3. Validate the user's input.

Ensure that:

- Every grade is between **0 and 100**.
- Only valid numeric values are accepted.

If the input is invalid, ask the teacher to enter it again.

4. Calculate the average.

Use the following formula:

$$\text{Average} = (\text{Math} + \text{Science} + \text{English}) \div 3$$

Round the average to **2 decimal places**.

5. Determine the student's result.

Display:

- Letter grade (A, B, C, D, or F)
- Pass or Fail

Passing score: 60 or above.

6. Display the student report.

Example:

```
=====
Student Report
=====
Student : Sara

Math : 95

Science : 88

English : 91

Average : 91.33

Letter Grade : A

Status : Passed
=====
```

7. Ask whether another student's grades should be calculated.

If the answer is:

- **Yes** → Start another calculation.
 - **No** → Display a goodbye message and end the application.
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Bonus Mission ★

After calculating the average, display an achievement message.

Average	Achievement
95 or above	Outstanding Student
90–94.99	Honor Student
80–89.99	Excellent Work
Below 80	Keep Improving

Useful Hints

- Use dictionaries for the grading scale.
 - Create functions for:
 - Calculating the average.
 - Determining the letter grade.
 - Determining pass/fail status.
 - Use while True to repeat the application.
 - Validate all grades before calculating.
 - Use round() for the average.
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GUI Mission (Streamlit)

Create a graphical version of the Student Grade Calculator.

Required Text Functions

Use:

- `st.title()`
 - `st.header()`
 - `st.subheader()`
 - `st.write()`
 - `st.divider()`
-

Required Input Widgets

Use:

- `st.text_input()` → Student name
 - `st.number_input()` → Math grade
 - `st.number_input()` → Science grade
 - `st.number_input()` → English grade
 - `st.button()` → Calculate Result
-

Required Status Elements

Use **at least three** of the following:

- `st.info()` → Explain the grading system.
 - `st.success()` → Student passed.
 - `st.warning()` → Average is between 60 and 69.
 - `st.error()` → Student failed.
-

Required Layout

Your application **must** include:

Sidebar

Display:

- Grading Scale
 - Passing Score
-

Two Columns

Left Column

- Student information
- Grades
- Calculate button

Right Column

Display:

- Average
 - Letter Grade
 - Pass/Fail
 - Achievement (Bonus)
-

Expected GUI

🎓 Student Grade Calculator

Sidebar

Grading Scale

A : 90–100

B : 80–89

C : 70–79

D : 60–69

F : Below 60

Passing Score : 60

Student Name

[Sara]

Math

[95]

Science

[88]

English

[91]

[Calculate Result]

Student Report

Average : 91.33

Grade : A

 Passed

Achievement

 Honor Student

Challenge Mission ★★

Allow the teacher to calculate the grades for **5 students**, then display:

- Highest average
- Lowest average
- Class average
- Number of students who passed
- Number of students who failed

Hint: Store each student's information in a **list of dictionaries**.

Python Topics Covered

- Variables
 - Strings
 - Integers
 - Floats
 - Dictionaries
 - Arithmetic operators
 - Comparison operators
 - Conditional statements (if, elif, else)
 - while loop
 - User-defined functions
 - Input validation
 - round()
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Streamlit Topics Covered

Displaying Text

- st.title()
- st.header()
- st.subheader()
- st.write()
- st.divider()

Input Widgets

- st.text_input()
- st.number_input()
- st.button()

Status Elements

- st.info()
- st.success()
- st.warning()
- st.error()

Layout

- `st.sidebar`
 - `st.columns()`
-

Learning Objectives

After completing this project, you should be able to:

- Validate user-entered grades.
- Calculate averages using arithmetic operations.
- Use conditional logic to classify grades and determine pass/fail status.
- Organize code into reusable functions.
- Design a user-friendly Streamlit application with a clear input section and a well-formatted report.